

Pitfalls of Single-Metric Evaluation for Quantized Reasoning Checkpoints

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Abstract

Practitioners often choose a public quantized reasoning checkpoint from a single published pass@1. We show three ways that single-metric choice misleads, using one pinned A100 stack (vLLM 0.7.0, eager [12]) and eight DeepSeek-R1-Distill checkpoints—Qwen-7B and Llama-8B in BF16, FP8 (executed as Marlin W8A16 [5]), community AWQ-4, and RedHatAI GPTQ-4—on MATH-500 [9, 15], GSM8K [2], and GPQA-Diamond [20] (88 runs, 56,408 completions).

(i) *Accuracy*. Only one drop survives item-level intervals and multiplicity control: Llama AWQ-4 (MATH-500 -2.76 pp; GSM8K -1.57 pp). Strict-majority voting over five samples (maj@5) halves the MATH gap to -1.4 pp. (ii) *Length*. The BF16-correct token-inflation contrast is biased by selection: comparing two BF16 seeds, with no quantization at all, already yields +138 (Qwen) and +235 (Llama) tokens, and conditioning on the quantized model’s success instead flips the sign in four of six contrasts. (iii) *Cost*. On a 20-prompt serving subset a single length draw decides the ranking: Llama FP8 averages 7,288 tokens there versus 4,551 on the full grid, and timing repeats that share one sampling seed cannot expose this. As a gold-free alternative to length heuristics, unanimous agreement of just two samples lowers selective risk to 0.7–2.4%, against 2.2–5.8% for a one-sample length rule at matched coverage; five samples reach at most 0.27%.

Keywords: Reasoning language models, quantization checkpoints, selection effects, majority vote, selective prediction.

1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Post-training quantization (FP8, AWQ-4 [16], GPTQ-4 [4]) is a common serving choice, and practitioners often pick a public checkpoint from a single published pass@1. Isolated pass@1 numbers hide whether the same checkpoint also changes trace length [14, 18, 19], termination, or the cost of a correct answer [3, 11]. Serving-stack details—engine version, decoding flags, CUDA graphs—can materially move reasoning scores [10]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment. The unit of analysis is the public checkpoint $\times$ pinned serving stack $\times$ evaluation target.

We ask four questions under that pin:

- **RQ1.** Which pass@1 differences survive an item-level interval, and how much of a 4-bit gap does maj@5 remove?

Table 1: Qwen-7B numbers reported by QRM beside this pinned grid. Cells are matched by benchmark and format. QRM quantized its own AWQ-4 model, whereas this grid evaluates the public `jakiAJK` upload, so the AWQ rows compare different artifacts. The GPQA row is not a certified paired contrast (Appendix Section A). The discrepancy is checkpoint- and stack-sensitive measurement, not a superiority claim.

Cell	QRM	This grid
MATH-500 BF16	94.60	94.00
MATH-500 AWQ-4	91.00	93.12
GPQA-Diamond AWQ-4	50.00	44.78

- **RQ2.** Is a BF16-correct length increase a property of the checkpoint, or a selection effect of conditioning on quantized failures?
- **RQ3.** At matched coverage, does k -sample answer agreement have lower selective risk than a one-sample length rule, and how does that gain scale with k ?
- **RQ4.** How much of a subset serving-cost gap is one length draw that timing repeats do not cover?

Contributions. Each contribution answers the research question with the same number. (C1) **One robust accuracy drop.** Llama AWQ-4 is the only contrast that is Holm-significant on both MATH-500 and GSM8K; `maj@5` recovers about half of it. AWQ results are properties of the `jakiAJK` artifacts run in `float16` on the GEMM `awq` kernel, not of the AWQ method. (C2) **A placebo-calibrated selection effect in the BF16-correct length estimator.** A BF16-versus-BF16 seed placebo reproduces a large part of the BF16-correct token increase without any quantization, and the mirrored quant-correct conditional flips sign in four of six contrasts. (C3) **A cost curve for gold-free agreement.** Unanimity of $k = 2\text{--}5$ samples has lower selective risk than a one-sample length rule at matched coverage at every k ; two samples already capture most of the reduction at twice the token cost. (C4) **Subset serving cost is a length draw.** Timing repeats share one sampling seed. Campaign-length seconds, using the same measured tok/s, put Llama FP8 first on both conditions in all 10,000 item resamples; the 20-prompt Condition A order does not.

2 Related Work

GPTQ [4] and AWQ [16] were originally validated mainly on short-form benchmarks. Liu et al. (QRM; COLM 2025) [18] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization, reporting accuracy and mean length. Kurtic et al. [11] study vLLM serving efficiency across formats and batching regimes. Lian et al. [14] report token inflation as an unconditional ratio of mean chain-of-thought lengths (their CTIR, the same estimand as our ratio-of-means column) and, in a supplementary correctness-conditioned analysis, find inflation stronger on the BF16-correct subset; RQ2 examines that conditional. Hochlehnert et al. [10] show that seeds, prompts, decoding, and software move reasoning scores. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution.

Existing studies evaluate quantization accuracy, throughput, or individual reasoning behaviors. Kurtic et al. already show that format recommendations change between synchronous decode and continuous batching. Our Condition A versus Condition B contrast is a pinned-stack sensitivity on these public checkpoints, not a second batching law. Appendix Table 22 reports mean tokens divided by `pass@1`. Appendix Table 23 places this grid against those studies. The unit of comparison is the public checkpoint $\times$ pinned stack $\times$ evaluation target.

Table 2: Activation dtype and vLLM kernel. Accuracy dtype is the 56k campaign. Serving dtype is the confirmation timer. AWQ uses the GEMM awq kernel, not awq-marlin. GPTQ’s checkpoint file is compressed-tensors; the kernel is Marlin W4A16. Accuracy-campaign GPTQ uses bfloat16 and the timer uses float16.

Format	Accuracy dtype	Accuracy kernel	Serving dtype
BF16	bfloat16	native	bfloat16
FP8	bfloat16	Marlin W8A16	bfloat16
AWQ-4	float16	GEMM awq	float16
GPTQ-4	bfloat16	Marlin W4A16	float16

The FP8 and GPTQ-4 cells are public RedHatAI compressed-tensors checkpoints evaluated as A100 Marlin weight-only kernels. These are not Red Hat production-deployment measurements. We do not rerun SmoothQuant [22]. Lotfi et al. [19], Li et al. [13], Helcig et al. [8], Zhong et al. [23], and Ayadi et al. [1] study overthinking, reasoning degradation, statistically lossless quantization, calibration, and reliability scaling; we do not re-estimate those quantities. Self-consistency [21] and G-Pass@ k [17] are different estimands from the unique-mode rule in Section 4.7. We report mean tokens divided by pass@1, and an aggregate hybrid Cost-of-Pass proxy from measured GPU-seconds; neither is Erol et al.’s per-problem estimator.

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM’s Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128); **GPTQ-4** (group size 128). Hugging Face IDs and revisions are in Appendix Table 17. Table 2 records the activation dtype and kernel. An AWQ-versus-BF16 contrast is therefore quantization plus a change from bfloat16 to float16 activations. Qwen-family models have a known float16 overflow history; this grid does not include a BF16 checkpoint rerun at dtype=float16, so that mechanism is not identified. Accuracy-campaign GPTQ-4 stays on bfloat16. The serving script sets float16 for both AWQ-4 and GPTQ-4, so the GPTQ accuracy cells and the GPTQ timing cells do not share an activation dtype.

3.2 Benchmarks and protocol

MATH-500 [9, 15] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [2] ($n = 1,319$, seeds 42–44) and GPQA-Diamond [20] ($n = 198$, seeds 42–44): breadth. Decoding follows the QRM chat template in Appendix Section A, with $T = 0.6$, top- $p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), and max new tokens 32,768. Total: 56,408 completions in 88 checkpoint×benchmark×seed runs. Throughout, a *run* denotes one checkpoint×benchmark×seed evaluation; a *cell* denotes one checkpoint×benchmark aggregated over its evaluated seeds; a *format contrast* compares a quantized checkpoint with its matched BF16 checkpoint within the same model family and benchmark. MATH and GSM8K are widely used; a paired within-item design reduces but does not eliminate contamination interaction with format.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM `inference.py` path [18] with `gpu_memory_utilization=0.75`. Engine pin:

vLLM==0.7.0, `enforce_eager=true`. Eager execution keeps the accuracy path reproducible and adds CPU kernel-launch overhead on single-stream decode. Production serving more often uses CUDA graphs. This paper has no CUDA-graph replicate, so Condition A format gaps are specific to eager execution. Effective launch settings are recorded in `results/reports/runtime_manifest.json`.

3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager. Primary numbers are the controlled confirmation (protocol: `docs/MEASURE_D_SERVING_CONFIRMATION_PROTOCOL.md`; raw JSON under `results/measured_serving_confirmation/raw/`). Condition A is 20 sequential MATH-500 prompts (4 per difficulty level). Condition B is 100 prompts (20 per level) in one `llm.generate` call with `max_num_seqs=8`. The two conditions differ in both the prompt subset and the serving regime, so a ranking change between them is a property of the whole condition. With `max_num_seqs=8`, Condition B elapsed time runs until the last trace finishes. One or two long traces can decode alone after the batch drains. The raw JSON stores aggregate tokens and elapsed time, not time spent below a full batch of 8, so that tail share is not reported. A secondary microbenchmark forces 10 prompts to 512 tokens (`ignore_eos`).

Three warmup queries are discarded. Δt_{steady} is then the wall-clock from the first measured request to the return of the last measured response, bracketed by `torch.cuda.synchronize()`. Engine initialization is outside that interval. The first measured prefill and the straggler tail are inside it. Output tok/s = $(\sum \text{output tokens}) / \Delta t_{\text{steady}}$. GPU-sec/query = $\Delta t_{\text{steady}} / N$ on one GPU. Sampling matches the campaign ($T = 0.6$, top- $p = 0.95$) with seed 20260816. Repetitions share that seed, so output tokens are identical across repeats. The tok/s spread is wall-clock jitter on one token sequence. It is not length variance. $R = 3$, expanded to $R = 5$ when the CV of the first three tok/s values exceeds 3%. With three repeats a percentile interval has at most ten distinct means. We therefore report the sample mean, the sample SD, and the observed min-max, and we do not treat those intervals as a stable estimate of timing noise. Qwen-7B jobs ran on `ragpu003` and Llama-8B jobs on `ragpu004`; both are two-GPU nodes, and each SLURM job requested one GPU and 16 CPU cores without `--exclusive`. The raw records log NVIDIA driver 550.54.14 on both hosts. Within a family, all four formats share a host but ran sequentially on one day (Qwen: BF16 05:36, FP8 08:29, AWQ-4 11:09, GPTQ-4 12:38 UTC), so format is also confounded with time on a node that other jobs could share. Eager decode is bound by kernel-launch CPU time, so host load moves single-stream tok/s. Across families, host and architecture are confounded. Archived logs do not contain per-run clocks, power, or co-tenant load. Peak VRAM is `torch.cuda.max_memory_allocated` after the 0.75 utilization preallocation.

$$\tilde{C}_{\text{pass}}^{\text{hyb}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \quad (1)$$

The numerator is subset GPU-seconds and the denominator is the frozen MATH-500 campaign `pass@1`. This ratio follows the Cost-of-Pass form and is not the per-problem estimator of Erol et al. [3]. Because the repeats share a seed, the printed interval multiplies wall-clock jitter by a `pass@1` bootstrap. It does not resample prompt lengths. A second estimator holds measured tok/s fixed and substitutes the campaign mean length: `campaign tokens ÷ tok/s ÷ pass@1`. That estimator still uses a length-dependent tok/s, so it is a sensitivity, not a replacement measurement. The \$1.50/A100-hour rate is a pricing scenario. Because it is a single constant, it rescales GPU-seconds without changing any ranking.

3.5 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is an item-level bootstrap of the paired difference $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$. For each problem i we first average correctness over seeds, $\delta_i = \overline{\text{acc}}_{i,\text{quant}} - \overline{\text{acc}}_{i,\text{BF16}}$, then resample n problems with replacement ($B = 10,000$; RNG seed 0). Seeds are averaged inside each item, then items are resampled. Resampling those item means carries the generation noise already averaged into them. The interval remains conditional on the seed set that was run. With bootstrap replicates Δ_b^* , the two-sided bootstrap tail-area p -value is

$$p_{\text{hi}} = \#\{\Delta_b^* \geq 0\}/B, \quad p_{\text{lo}} = \#\{\Delta_b^* \leq 0\}/B, \quad p = \min(1, 2 \min(p_{\text{hi}}, p_{\text{lo}})). \quad (2)$$

This is the p -value obtained by inverting the percentile interval. A reported $p = 0$ or $p < 0.001$ means that no sign-crossing bootstrap replicate was observed among the 10,000 draws; it is not a literal probability of zero. When a Holm decision lies within Monte Carlo error of α , we rerun that contrast with $B = 10^6$ and also report an item-level sign-flip permutation test (2×10^5 sign vectors) and a paired t -test on the item means. Holm–Bonferroni is applied separately to the six primary within-benchmark format contrasts (MATH-500, GSM8K, GPQA-Diamond). Each benchmark is a distinct inferential family because each corresponds to a separate task distribution; we do not pool the 18 contrasts for the primary analysis. A joint Holm correction over all 18 contrasts is reported only as a stricter sensitivity (Appendix Table 24). Two one-sided tests (TOST) at a ± 1 pp equivalence margin use the 90% CI; this is an equivalence check, not a default claim. We use ± 1 pp as a study tolerance, not a universal equivalence standard. On MATH-500 the paired standard error is about 0.45 pp, so a ± 1 pp TOST passes only when the point estimate is close to zero. The smallest margin that would contain the FP8 90% intervals is 1.12 pp for Qwen and 1.28 pp for Llama. A failed ± 1 pp test at this n reflects the sample size relative to that margin.

maj@5. A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold, i.e. if a strict majority of the five answers is correct. This is strict-majority voting; it is a lower bound on plurality voting, which can also be correct with two correct answers and three mutually different wrong ones. We report the accuracy, not only the McNemar test against BF16. Non-significant maj@5 tests do not imply that pass@1 effects are absent.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). The launcher sets both `max_model_len` and `max_new_tokens` to 32,768. In vLLM the first bound is prompt plus output, so the effective output cap is 32,768 minus that item’s prompt length. Compact JSON has no `finish_reason`. Under each family’s chat template MATH-500 prompts span 29–811 tokens (Qwen) and 27–776 (Llama), so the effective cap lies between about 31,957 and 32,741 tokens. We therefore report *near-cap* rows ($\geq 32,500$ tokens) as the truncation proxy. A per-item truncation count needs each row’s own prompt; the compact JSON stores items in the campaign pipeline order, which is not the order of the public dataset release, so that count is deferred to the campaign prompt files (Section 4.3). The `boxed` flag is a string search for `\boxed`, which is separate from extractive match.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with an item-level percentile bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness, with item-level 95% CIs on each cell. A direct contrast $D = \overline{\Delta}_{\text{BF16-only}} - \overline{\Delta}_{\text{Both-OK}}$ uses the same item resamples for both stratum means; replicates in which either stratum is empty are skipped. D compares conditional means and is not a causal effect. A BF16-internal control compares BF16 length when BF16 is correct versus

Table 3: MATH-500 headline grid ($n = 500$, seeds 42–46). Item-level 95% CIs resample problems. maj@5: items with at least 3 of 5 seeds correct. Loops: identical-word run ≥ 20 . NC: near-cap, completion tokens $\geq 32,500$, the truncation proxy (Section 4.3).

Cell	42	43	44	45	46	Mean $\pm$ Std	Item 95%	Tok.	maj@5	L	NC
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	94.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	95.0	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	94.4	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	94.0	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	91.2	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	91.0	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	89.8	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	91.2	2	21

incorrect. The BF16-correct conditional Δ , following Lian et al., is the mean quantized–BF16 Δ on pairs where BF16 is correct (Both-OK $\cup$ BF16-only). The mirrored quant-correct conditional is the same mean on pairs where the quantized checkpoint is correct (Both-OK $\cup$ Quant-only). Conditioning on BF16-correct pulls in quantized failures, which are long, so the BF16-correct column is positive even when the unconditional change is near zero. Conditioning on quant-correct pulls in BF16 failures and moves the sign the other way. Both columns are selection effects. To size the selection effect directly we run a *seed placebo*: the same BF16-correct estimator applied to BF16 seed s against BF16 seed $t \neq s$, pooled over all 20 ordered seed pairs, with an item-level bootstrap CI. Any nonzero placebo value is produced by conditioning alone, because both arms are the same checkpoint. D compares the BF16-only and Both-OK means on shared item resamples and is a mismatch diagnostic, not a causal token increase.

Tokens per correct answer. Appendix Table 22 reports campaign mean tokens divided by MATH-500 pass@1. That ratio is what a shared throughput would rank. It does not use a tok/s constant. Primary cost evidence is Table 14 together with the campaign-length sensitivity in Table 15. The hybrid dollar in Table 14 divides measured GPU-seconds by pass@1. For GPTQ-4 those GPU-seconds were timed at `float16`, while the pass@1 denominator comes from the `bfloat16` accuracy campaign. The two GPTQ cells are not the same configuration.

Observable modal agreement. Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement without gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). For item i and five generated predictions, let $\sim$ be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of $\sim$. Let m_i be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is $a_i = m_i/5$. Serve at threshold $k/5$ only when a unique modal class satisfies $m_i \geq k$. The pattern (A, A, B, B, C) has $m_i = 2$ and is not served at $\geq 3/5$; (A, A, A, B, B) has unique mode A with $m_i = 3$ and is served at $\geq 3/5$. Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy T_5 is the sum of their completion tokens, including abstained queries. We report three discrete operating points, not a continuous risk–coverage curve. Coverage and selective risk follow the selective-classification framework of Geifman and El-Yaniv [6]. For the cost curve we also use a k -sample unanimity rule for $k = 2, \dots, 5$: serve when all k answers fall in one equivalence class, and average coverage and risk over all $\binom{5}{k}$ seed subsets. Its token cost is k times the mean completion length. Gold-hit ECE (confidence = $c_i/5$, label = $\mathbb{I}(c_i \geq 3)$) remains a deterministic function of the gold-hit histogram and is omitted.

Table 4: Primary: item-level bootstrap of MATH-500 pass@1 versus BF16 ($B = 10,000$). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts (primary family). TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST	McNemar p
Llama AWQ-4	-2.76	[-4.16, -1.44]	< 0.001	sig.	[-3.92, -1.64]	fail	0.296
Qwen AWQ-4	-0.88	[-1.88, +0.08]	0.069	n.s.	[-1.68, -0.08]	fail	1.000
Qwen GPTQ-4	-0.52	[-1.40, +0.36]	0.245	n.s.	[-1.28, +0.20]	fail	0.754
Qwen FP8	+0.40	[-0.48, +1.28]	0.383	n.s.	[-0.32, +1.12]	fail	0.581
Llama GPTQ-4	-0.32	[-1.56, +0.96]	0.619	n.s.	[-1.36, +0.72]	fail	1.000
Llama FP8	+0.28	[-0.92, +1.48]	0.647	n.s.	[-0.72, +1.28]	fail	1.000

4 Results

4.1 MATH-500 pass@1

Table 3 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. Llama AWQ-4 has the largest MATH-500 pass@1 drop among the Llama cells (−2.76 pp).

Table 4 is the test that matches RQ1. Llama AWQ-4 is the only MATH contrast whose interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). The other five MATH pass@1 contrasts are not significant, so a strict pass@1 order among the Qwen cells is not identified. An item bootstrap ($B = 10,000$) puts Qwen FP8 first on pass@1 in 80.3% of resamples and Llama FP8 first in 60.4%; Llama AWQ-4 is never first. No MATH contrast has its 90% CI inside $[-1, +1]$ pp. maj@5 accuracies are in Table 3: Qwen 94.4 / 95.0 / 94.4 / 94.0 and Llama 91.2 / 91.0 / 89.8 / 91.2 for BF16 / FP8 / AWQ-4 / GPTQ-4. Majority voting removes Qwen AWQ-4’s −0.88 pp pass@1 gap (both 472/500, McNemar $p = 1$) and about half of Llama AWQ-4’s gap (−2.76 to −1.4 pp). It also swaps Qwen AWQ-4 and GPTQ-4 relative to pass@1. McNemar p -values on maj@5 are all > 0.29 .

4.2 GSM8K and GPQA-Diamond

Table 10 gives cell means; Tables 5 and 6 are the item-level tests. FP8 stays within about one point of BF16 on all three tasks. Several GSM8K contrasts pass TOST at ± 1 pp, where $n = 1,319$ and the interval is narrower than on MATH-500. Llama AWQ-4 also drops on GSM8K (−1.57 pp, 95% CI [−2.55, −0.58], $p = 0.002$). Qwen AWQ-4 differs from Qwen BF16 by −5.56 pp on GPQA-Diamond (95% CI [−9.60, −1.52]). Its significance is borderline. With $B = 10,000$ the bootstrap p ranged from 0.0056 to 0.0104 across 20 resampling seeds, which puts the Holm-6 decision on both sides of α . With $B = 10^6$ the bootstrap p is 0.0081 (Holm-6 adjusted 0.049). The sign-flip permutation test gives $p = 0.011$ (adjusted 0.066) and the paired t -test gives $p = 0.009$ (adjusted 0.054). We therefore label it borderline rather than significant. It is not significant under the Holm-18 sensitivity (adjusted $p = 0.130$; Appendix Table 24). Appendix Section A shows that QRM draws the GPQA correct letter with `random.randint` after model load, so this contrast is not a certified paired comparison. It is not a headline result.

4.3 Loops and near-cap completions

Across 88 runs the validator records **25** repetition-flagged completions (0.044% of 56,408). The observed maximum completion length is 32,737. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ). Three completions exceed 10,000 identical consecutive words. Near-cap completions ($\geq 32,500$ tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16’s 14.

Table 5: GSM8K item-level pass@1 versus matched BF16 ($n = 1,319$, 3 seeds). Holm over six contrasts. TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST
Llama AWQ-4	-1.57	[-2.55, -0.58]	0.002	sig.	[-2.38, -0.73]	fail
Qwen AWQ-4	-0.20	[-0.91, +0.48]	0.579	n.s.	[-0.78, +0.38]	pass
Qwen GPTQ-4	-0.13	[-0.86, +0.63]	0.729	n.s.	[-0.76, +0.51]	pass
Llama GPTQ-4	+0.28	[-0.53, +1.09]	0.511	n.s.	[-0.40, +0.96]	pass
Qwen FP8	+0.08	[-0.58, +0.73]	0.827	n.s.	[-0.45, +0.63]	pass
Llama FP8	+0.13	[-0.66, +0.94]	0.761	n.s.	[-0.53, +0.81]	pass

Table 6: GPQA-Diamond pass@1 versus matched BF16 ($n = 198$, 3 seeds). The rows are descriptive. Choice order is not certified as paired across checkpoints (Appendix Section A). The Qwen AWQ-4 p uses $B = 10^6$ (other rows $B = 10,000$); it is borderline under Holm-6 (adjusted 0.049 bootstrap, 0.066 permutation) and not significant under Holm-18, even if the pairing is taken at face value. It is not a headline claim. border.: borderline; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST
Qwen AWQ-4	-5.56	[-9.60, -1.52]	0.008	border.	[-9.09, -2.19]	fail
Qwen GPTQ-4	-2.36	[-6.57, +1.85]	0.288	n.s.	[-6.06, +1.18]	fail
Llama GPTQ-4	-1.18	[-5.39, +3.03]	0.573	n.s.	[-4.71, +2.36]	fail
Qwen FP8	-0.84	[-4.71, +3.03]	0.690	n.s.	[-4.21, +2.53]	fail
Llama AWQ-4	+0.84	[-3.54, +5.05]	0.711	n.s.	[-2.86, +4.38]	fail
Llama FP8	+1.68	[-2.19, +5.39]	0.403	n.s.	[-1.52, +4.88]	fail

Near-cap rows are very likely truncations. Of the 140 MATH-500 near-cap rows, 121 lie between 32,600 and 32,739 tokens, the band where 32,768 minus a typical prompt falls. Of the 57 GSM8K near-cap rows, 55 lie in that band. Only 24 MATH-500 rows and no GSM8K rows fall between the lowest possible cap (31,957) and 32,500. An exact per-item truncation count cannot be obtained by joining public prompt lengths on row position, because the compact JSON stores items in the campaign pipeline order, not the public release order. It requires the prompt text stored in the campaign output files. On MATH-500, 120 of 134 unboxed rows are near-cap. A missing box on MATH is usually a long trace. It is not the 4-bit accuracy gap: unboxed rows that are still scored correct are rare (at most two per seed file), and boxed rates are 98.96–99.68%. GSM8K is similar (52 of 66 unboxed rows are near-cap; boxed rates 99.70–99.87%). GPQA is different: 4,746 of 4,752 rows are unboxed, and only 12 are near-cap. GPQA extractive match is not a boxed-answer grade, and the long-tail proxy does not explain it. Native stop reasons were not logged, so near-cap is a length proxy, not a `finish_reason` count.

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 8, Figure 3), the tested Qwen AWQ-4/GPTQ-4 checkpoints showed +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Excluding near-cap pairs ($\geq 32,500$ tokens) the ratios remain +4.61%/+5.75%. Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A 200-item even-index, seed-42, mean-of-ratios analysis is an earlier estimator. It is not the full-grid ratio of means. The note is in the repository changelog, not in these results.

Figure 4 and Table 8 split paired token deltas by the 2×2 . Qwen AWQ-4 and GPTQ-4 Both-OK intervals exclude zero (+166 [+80, +254] and +136 [+45, +239]). Qwen FP8 and all three Llama Both-OK intervals include zero. The BF16-correct conditional Δ is positive with

Table 7: Seed placebo for the correctness-conditioned length estimator on MATH-500. “Reference” is the arm whose correctness is conditioned on. Placebo rows compare BF16 seed s with BF16 seed $t \neq s$, pooled over 20 ordered pairs. Quantized rows repeat the BF16-correct column of Table 8. Item-level 95% CIs, $B = 10,000$.

Family	Contrast	Reference-correct Δ [95% CI]
Qwen-7B	BF16 vs BF16 (placebo)	+138 [+75, +207]
Qwen-7B	FP8 vs BF16	+172 [+63, +291]
Qwen-7B	AWQ-4 vs BF16	+390 [+261, +531]
Qwen-7B	GPTQ-4 vs BF16	+326 [+198, +464]
Llama-8B	BF16 vs BF16 (placebo)	+235 [+151, +331]
Llama-8B	FP8 vs BF16	+166 [+33, +302]
Llama-8B	AWQ-4 vs BF16	+332 [+182, +490]
Llama-8B	GPTQ-4 vs BF16	+417 [+261, +583]

Table 8: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ : mean paired per-item token difference with an item-level 95% CI. 2×2 cells are mean quantized-BF16 Δ with an item-level 95% CI and pair count n . BF16-correct and quant-correct columns use the same item resamples on Both-OK $\cup$ BF16-only and Both-OK $\cup$ Quant-only.

Contrast	RoM	Mean Δ [95% CI]	Both OK	BF16 only	Quant only	Both wrong	BF16-correct	Quant-correct
Qwen FP8	-0.1%	-4 [-142, +134]	+51 [-42, +145] ($n=2297$)	+5444 [+2922, +8035] ($n=53$)	-6940 [-9479, -4345] ($n=63$)	+264 [-1126, +1685] ($n=87$)	+172 [+63, +291]	-136 [-273, -3]
Qwen AWQ-4	+6.3%	+254 [+116, +398]	+166 [+80, +254] ($n=2272$)	+6915 [+4534, +9345] ($n=78$)	-6533 [-8933, -4255] ($n=56$)	+894 [-620, +2420] ($n=94$)	+390 [+261, +531]	+5 [-111, +114]
Qwen GPTQ-4	+6.9%	+276 [+121, +431]	+136 [+45, +239] ($n=2284$)	+6892 [+4090, +9896] ($n=66$)	-4910 [-8180, -1715] ($n=53$)	+1893 [+204, +3588] ($n=97$)	+326 [+198, +464]	+22 [-99, +151]
Llama FP8	-2.3%	-106 [-284, +65]	+38 [-74, +152] ($n=2121$)	+2629 [+1098, +4265] ($n=110$)	-4159 [-6040, -2428] ($n=117$)	-968 [-2252, +198] ($n=152$)	+166 [+33, +302]	-182 [-339, -35]
Llama AWQ-4	+1.7%	+80 [-109, +268]	+91 [-25, +210] ($n=2052$)	+3090 [+1995, +4286] ($n=179$)	-4477 [-6503, -2588] ($n=110$)	-305 [-1520, +907] ($n=159$)	+332 [+182, +490]	-141 [-303, +12]
Llama GPTQ-4	+3.9%	+184 [+12, +359]	+66 [-43, +181] ($n=2097$)	+5915 [+4335, +7605] ($n=134$)	-4327 [-6138, -2561] ($n=126$)	+521 [-506, +1585] ($n=143$)	+417 [+261, +583]	-183 [-335, -35]

intervals excluding zero in every contrast (+166 to +417 tokens), including Qwen FP8, whose unconditional change is -4 tokens $[-142, +134]$. That column equals the n -weighted mix of Both-OK and BF16-only. For Qwen FP8, $(2297 \times 51 + 53 \times 5444)/2350 = 172$. Failure traces are long (15,843 vs. 3,256 tokens on Qwen BF16; 13,928 vs. 3,539 on Llama). The mirrored quant-correct conditional, the n -weighted mix of Both-OK and Quant-only, is $-136 [-273, -3]$, $+5 [-111, +114]$, $+22 [-99, +151]$, $-182 [-339, -35]$, $-141 [-303, +12]$, and $-183 [-335, -35]$ tokens. Three of those six intervals exclude zero, all on the negative side. Qwen AWQ-4 and GPTQ-4 stay near zero because the Both-OK lengthening offsets a smaller Quant-only tail.

The seed placebo (Table 7) sizes this selection effect directly. Applying the BF16-correct estimator to two BF16 seeds, with no quantization, gives +138 [+75, +207] tokens for Qwen and +235 [+151, +331] for Llama. The mirrored conditional is exactly the negative, by symmetry. Both FP8 BF16-correct values (+172 and +166) lie inside their family’s placebo interval, so the whole FP8 “inflation” under this estimator is reproduced by conditioning alone. The four 4-bit values (+326 to +417) exceed their placebo by about 100–250 tokens. That excess, not the raw conditional, is the part a quantization effect could explain. It is close to the unconditional paired Δ (Qwen AWQ-4 +252 vs. +254; Llama GPTQ-4 +182 vs. +184; Llama AWQ-4 +97 vs. +80). Same-seed pairing does not couple the two arms: correctness agreement is 0.9536 for same-seed Qwen FP8/BF16 pairs and 0.9526 across seeds, with similar gaps in every contrast. The seed-matched 2×2 is therefore one arbitrary pairing, not a coupled design. We report both conditionals and the placebo, and we do not call any of them a reconciliation of prior length claims. D (Table 9) remains a mismatch diagnostic.

Mean tokens divided by pass@1 (Appendix Table 22) ranks Qwen FP8 first among Qwen cells. Campaign-length seconds, which divide by the measured tok/s, do not (Section 4.8).

Table 9: Item-level mismatch excess $D = \bar{\Delta}_{\text{BF16-only}} - \bar{\Delta}_{\text{Both-OK}}$ on MATH-500. Both stratum means are formed on the same item resamples ($B = 10,000$; empty-stratum replicates skipped; none occurred). D is a diagnostic of correctness-conditioned mismatch asymmetry, not a causal quantization-induced token increase.

Contrast	Both-OK Δ	BF16-only Δ	Mismatch excess D [95% CI]
Qwen FP8	+51 [-42, +145]	+5444 [+2922, +8035]	+5394 [+2853, +7992]
Qwen AWQ-4	+166 [+80, +254]	+6915 [+4534, +9345]	+6748 [+4375, +9184]
Qwen GPTQ-4	+136 [+45, +239]	+6892 [+4090, +9896]	+6755 [+3938, +9786]
Llama FP8	+38 [-74, +152]	+2629 [+1098, +4265]	+2591 [+1047, +4244]
Llama AWQ-4	+91 [-25, +210]	+3090 [+1995, +4286]	+2999 [+1898, +4187]
Llama GPTQ-4	+66 [-43, +181]	+5915 [+4335, +7605]	+5849 [+4258, +7553]

Table 10: Cross-benchmark pass@1 (mean $\pm$ seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

Cell	MATH-500	GSM8K	GPQA-D	L / NC
Qwen BF16	94.00 $\pm$ 0.55	91.26 $\pm$ 0.29	50.34 $\pm$ 2.96	4 / 28
Qwen FP8	94.40 $\pm$ 1.05	91.33 $\pm$ 0.16	49.49 $\pm$ 1.52	0 / 24
Qwen AWQ-4	93.12 $\pm$ 0.67	91.05 $\pm$ 1.14	44.78 $\pm$ 3.04	5 / 32
Qwen GPTQ-4	93.48 $\pm$ 0.77	91.13 $\pm$ 0.27	47.98 $\pm$ 1.75	3 / 40
Llama BF16	89.24 $\pm$ 0.74	88.68 $\pm$ 0.46	46.13 $\pm$ 1.91	3 / 21
Llama FP8	89.52 $\pm$ 1.01	88.80 $\pm$ 0.62	47.81 $\pm$ 0.29	3 / 12
Llama AWQ-4	86.48 $\pm$ 1.96	87.11 $\pm$ 0.23	46.97 $\pm$ 2.02	4 / 21
Llama GPTQ-4	88.92 $\pm$ 1.55	88.96 $\pm$ 0.70	44.95 $\pm$ 4.32	3 / 31

4.5 Difficulty labels

A per-level breakdown needs each row’s MATH-500 `level1`. The compact JSON stores items in the campaign pipeline order, which differs from the public release order: among correct rows with integer gold answers, the recovered prediction equals the public gold at the same position in only 118 of 11,382 cases. A positional join would therefore assign the wrong level to almost every row, so we do not report a per-level breakdown. The serving subsets are unaffected, because they were drawn by matching prompt text (Section 3.4).

4.6 Item-level descriptive checks

CPU-only reanalysis of the released compact JSON (`results/reports/item_level_descriptive_report.md`) is descriptive, not causal. MATH-500 has no item that is BF16-correct and quantized-wrong on all five seeds for any tested format. Incorrect MATH traces are longer than correct traces in every cell (for example, Qwen BF16 mean tokens 3,256 when correct vs. 15,843 when incorrect). GPQA-Diamond Qwen AWQ flips remain distributed (75 of 198 items flip on at least one seed; 0 items flip on all three seeds). These counts do not identify a small set of items that would explain the GPQA contrast.

4.7 Observable modal agreement on MATH-500

Because MATH-500 was evaluated with five generations per item, we additionally examine whether observable answer agreement provides a useful gold-free abstention signal. Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration, and not as G-Pass@ k [17]. Table 13 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher

Table 11: Cost curve for k -sample unanimity on MATH-500. Coverage (Cov.) and selective risk (Risk) are averaged over all $\binom{5}{k}$ seed subsets. Len.: one-sample shortest-trace rule at the same coverage, averaged over five seeds. Tokens per item are k times the mean completion length. The $k = 5$ column equals the 5/5 column of Tables 13 and 12. All values are percentages except tokens.

Model	Fmt	$k = 2$			$k = 3$			$k = 4$			$k = 5$		
		Cov.	Risk	Len.	Cov.	Risk	Len.	Cov.	Risk	Len.	Cov.	Risk	Len.
Qwen-7B	BF16	92.5	0.89	2.20	90.7	0.44	1.94	89.4	0.27	1.83	88.4	0.23	1.67
Qwen-7B	FP8	92.7	0.69	2.16	90.7	0.18	1.94	89.6	0.00	1.88	88.8	0.00	1.71
Qwen-7B	AWQ-4	91.2	1.05	2.54	88.8	0.43	2.34	87.4	0.27	2.15	86.4	0.23	1.99
Qwen-7B	GPTQ-4	91.6	0.87	2.31	89.3	0.29	1.92	87.8	0.09	1.69	86.8	0.00	1.62
Llama-8B	BF16	85.8	1.86	4.06	81.5	0.71	3.34	78.5	0.36	3.16	76.2	0.26	2.83
Llama-8B	FP8	86.9	1.91	4.47	82.9	0.63	3.82	80.3	0.20	3.39	78.2	0.00	3.17
Llama-8B	AWQ-4	82.1	2.36	5.84	76.8	1.07	4.90	73.1	0.44	4.71	70.2	0.00	4.56
Llama-8B	GPTQ-4	85.1	1.76	4.08	80.5	0.62	3.03	77.6	0.31	2.78	75.4	0.27	2.55
Tokens / item (Qwen; Llama)		$\approx 8.0\text{--}8.6\text{k}; 9.1\text{--}9.7\text{k}$			$\approx 12.0\text{--}12.9\text{k}; 13.7\text{--}14.5\text{k}$			$\approx 16.0\text{--}17.1\text{k}; 18.2\text{--}19.4\text{k}$			$\approx 20.0\text{--}21.4\text{k}; 22.8\text{--}24.2\text{k}$		

observable answer agreement is associated with lower selective error. At 5/5 consensus, observed selective error is at most 0.27% across the evaluated configurations, but coverage ranges from 70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Several 0.00% risk cells have bootstrap intervals $[0, 0]$; Wilson 95% upper bounds on those 0/ n cells are 0.82%–1.08%. Zero observed errors is not zero true error.

Llama AWQ-4 5/5 coverage is 70.2% versus 76.2% for BF16 (-6.0 pp, 95% CI $[-9.4, -2.6]$). Two further coverage intervals exclude zero and stop at -0.6 pp: Llama AWQ-4 at $\geq 4/5$ (-3.2 pp) and Qwen AWQ-4 at $\geq 4/5$ (-2.2 pp). The remaining coverage contrasts include zero. These 18 intervals are exploratory. They do not receive the Holm correction used for pass@1. Among 18 tests at $\alpha = 0.05$, about one false positive is expected. The Llama AWQ-4 5/5 interval is the one that sits well clear of zero. Appendix Table 18 lists the full pattern.

A one-sample length rule is also gold-free and costs one generation. For each seed we serve the shortest k traces, with k set to the published 5/5 coverage of that cell, and average the selective risk over the five seeds (Table 12). At that matched coverage the length rule’s risk is 1.61–4.56%, against at most 0.27% for 5/5 modal agreement. The five-sample rule still has the lower selective risk. It also spends about five times the tokens (T_5 below).

Table 11 gives the cost curve between those two ends. Unanimity of only two samples serves 82.1–92.7% of items at 0.69–2.36% risk. The length rule at the same coverage has 2.16–5.84% risk, so two samples cut risk by a factor of about 2.2–3.1 for twice the tokens. Three samples reach 0.18–1.07%. Every k beats the matched length rule in every cell. Most of the gain comes from the second sample, so a deployment that cannot afford five generations keeps most of the benefit at $k = 2$ or 3. These values come from the recovered answer strings, with equivalence recomputed by math-verify on the stored representations. The same code reproduces every 5/5 coverage, 5/5 risk, and length-rule entry in Tables 13 and 12 exactly.

Mean T_5 is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. T_5 is five times the mean completion length. Abstained items still pay those tokens. Figure 1 shows the three operating points.

4.8 Measured serving throughput and Cost-of-Pass

Table 14 separates tok/s from GPU-seconds per query. GPU-seconds equal output tokens divided by tok/s, so a long subset draw raises cost even when the kernel is faster. Multiplying the printed tok/s by GPU-s/q recovers the subset length. Llama FP8 Condition A is about 7,288 tokens per query. The same checkpoint’s campaign mean is 4,550.8, which is 2.3% shorter than Llama BF16. That single 20-prompt draw, not a slower kernel, is why Condition A ranks

Table 12: One-sample length abstention at the published 5/5 coverage. For each seed, serve the shortest k traces. Risk is the mean over seeds 42–46. Modal risk is the 5/5 column of Table 13.

Model	Fmt	k	Length risk (%)	5/5 modal risk (%)
Qwen-7B	BF16	442	1.67	0.23
Qwen-7B	FP8	444	1.71	0.00
Qwen-7B	AWQ-4	432	1.99	0.23
Qwen-7B	GPTQ-4	434	1.61	0.00
Llama-8B	BF16	381	2.83	0.26
Llama-8B	FP8	391	3.17	0.00
Llama-8B	AWQ-4	351	4.56	0.00
Llama-8B	GPTQ-4	377	2.55	0.27

Table 13: Observable modal-answer agreement on MATH-500 ($n = 500$, seeds 42–46). Coverage is the fraction of items with a unique modal class of size $\geq k$. Risk is 1–selective accuracy among served items. Mean T_5 is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

Model	Fmt	$\geq 3/5$ cov.	$\geq 3/5$ risk	$\geq 4/5$ cov.	$\geq 4/5$ risk	5/5 cov.	5/5 risk	Mean T_5
Qwen-7B	BF16	96.0	1.67	93.4	0.43	88.4	0.23	20,057
Qwen-7B	FP8	96.6	1.66	92.6	0.00	88.8	0.00	20,038
Qwen-7B	AWQ-4	95.8	1.46	91.2	0.44	86.4	0.23	21,327
Qwen-7B	GPTQ-4	95.4	1.47	92.8	0.43	86.8	0.00	21,436
Llama-8B	BF16	94.0	2.98	87.8	0.68	76.2	0.26	23,283
Llama-8B	FP8	94.0	3.19	88.6	0.90	78.2	0.00	22,754
Llama-8B	AWQ-4	93.2	3.65	84.6	1.89	70.2	0.00	23,682
Llama-8B	GPTQ-4	93.8	2.77	86.4	0.46	75.4	0.27	24,203

Llama BF16 cheapest (\$0.0310 vs. about \$0.043). Llama GPTQ-4 Condition B tok/s is 0.2% below BF16 (366.16 vs. 366.98) while hybrid cost is +24.9%; the gap is mostly length.

The brackets in Table 14 are the minimum and maximum hybrid cost across timing repeats, with campaign pass@1 held fixed. They are not a 95% interval. Repeats share seed 20260816, so they do not include length variance. Qwen BF16 Condition A tok/s runs only from 43.88 to 43.96. Qwen FP8 Condition B runs from 350.68 to 546.26, and its dollar min–max is \$0.0030–\$0.0047. Qwen FP8 Condition B tok/s of 449.79 is the mean of five ratios. Mean GPU-s/q is 8.64, and $3,738/8.64 \approx 432.5$ tok/s. The two aggregations are not the same. Appendix Table 21 lists the five repeats. Llama FP8 and AWQ-4 Condition A means both round to \$0.0431. Their repeat ranges do not: \$0.0421–\$0.0449 and \$0.0423–\$0.0463.

Table 15 substitutes campaign mean tokens for the subset length and keeps the measured tok/s. Under that sensitivity, Qwen GPTQ-4 is cheapest on Condition A and tied with FP8 on Condition B, so the AWQ/GPTQ reversal between A and B disappears. Llama FP8 is cheapest on both conditions and on mean tokens divided by pass@1. An item bootstrap of campaign tokens and pass@1, with tok/s held fixed ($B = 10,000$), gives Llama FP8 rank 1 on both campaign-length costs in all 10,000 draws. Qwen GPTQ-4 is rank 1 on Condition A in 99.9% of draws and on Condition B in 59.6%, with FP8 taking the other 40.4%. These resamples hold tok/s fixed, so they carry no timing uncertainty. The Qwen FP8/GPTQ-4 Condition B split in particular lies inside the FP8 repeat range. Measured tok/s still depends on the subset’s length, so Table 15 is not a final serving measurement. Figure 2 plots the Condition B campaign-length seconds, not the subset dollars.

Qwen BF16 decodes at 43.92 tok/s on Condition A and 205.41 tok/s in the fixed-512 microbenchmark. Llama BF16, on the other host, reaches 72.34 and 362.01. The fixed-length gap remains, so it is not the Condition A length draw. The Qwen BF16 baseline looks anoma-

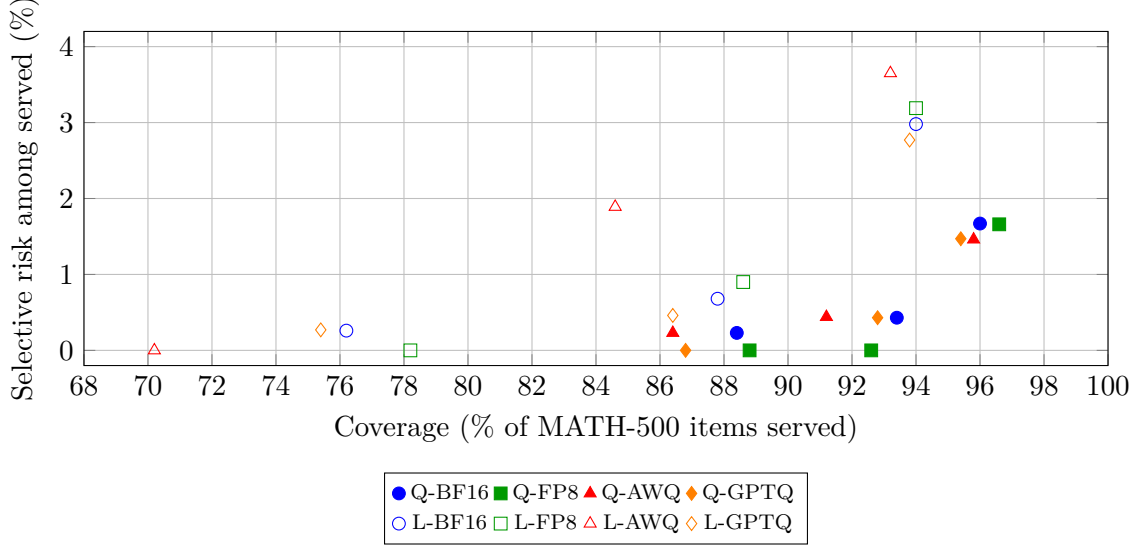


Figure 1: Three discrete MATH-500 operating points ($\geq 3/5$, $\geq 4/5$, $5/5$), not a continuous risk-coverage curve. Filled markers: Qwen-7B. Open markers: Llama-8B. Lower-right is high coverage and higher residual risk; lower-left is strict consensus.

Table 14: Hybrid cost on the controlled confirmation. Condition A: 20 sequential prompts. Condition B: 100-prompt `llm.generate, max_num_seqs=8`. Tok/s is mean $\pm$ sample SD. Brackets are the minimum and maximum hybrid cost across timing repeats, with campaign pass@1 held fixed. They are not a 95% interval. The B $\Delta\%$ column is the point estimate only. Qwen: `ragpu003`. Llama: `ragpu004`. GPTQ dollars join `float16` timing with `bfloat16` pass@1.

Model	Fmt	A tok/s	A GPU-s/q	A $\bar{C}$ [min, max]	B tok/s	B GPU-s/q	B $\bar{C}$ [min, max]	B $\Delta\%$
Qwen-7B	BF16	43.92 $\pm$ 0.04	102.51	\$0.0454 [0.0454, 0.0455]	252.72 $\pm$ 0.56	13.44	\$0.0060 [0.0059, 0.0060]	anchor
Qwen-7B	FP8	62.50 $\pm$ 0.08	84.60	\$0.0373 [0.0373, 0.0374]	449.79 $\pm$ 97.53	8.64	\$0.0038 [0.0030, 0.0047]	-36.0
Qwen-7B	AWQ-4	76.89 $\pm$ 0.70	48.04	\$0.0215 [0.0213, 0.0217]	418.15 $\pm$ 2.03	7.82	\$0.0035 [0.0035, 0.0035]	-41.3
Qwen-7B	GPTQ-4	82.67 $\pm$ 0.69	54.39	\$0.0242 [0.0241, 0.0245]	488.26 $\pm$ 0.64	7.23	\$0.0032 [0.0032, 0.0032]	-45.9
Llama-8B	BF16	72.34 $\pm$ 0.27	66.43	\$0.0310 [0.0309, 0.0311]	366.98 $\pm$ 1.90	10.13	\$0.0047 [0.0047, 0.0048]	anchor
Llama-8B	FP8	78.75 $\pm$ 2.89	92.55	\$0.0431 [0.0421, 0.0449]	481.42 $\pm$ 1.36	9.32	\$0.0043 [0.0043, 0.0044]	-8.2
Llama-8B	AWQ-4	70.20 $\pm$ 2.71	89.51	\$0.0431 [0.0423, 0.0463]	391.11 $\pm$ 0.48	11.63	\$0.0056 [0.0056, 0.0056]	+18.5
Llama-8B	GPTQ-4	63.49 $\pm$ 0.41	86.95	\$0.0407 [0.0405, 0.0410]	366.16 $\pm$ 0.79	12.60	\$0.0059 [0.0059, 0.0059]	+24.9

lous. Qwen-7B is the smaller model, yet it decodes about 40% slower than Llama-8B BF16. In the fixed-length microbenchmark FP8 speeds up Qwen by $2.13\times$ ($437.19/205.41$) but Llama by only $1.08\times$ ($392.05/362.01$), even though both use the same Marlin W8A16 kernel. Every Qwen saving in Table 14 (-36.0% to -45.9%) is relative to that baseline, which ran first on `ragpu003`. The Qwen FP8 Condition B repeats stepped up in run order (350.9, 350.7, 455.9, 545.2, 546.3 tok/s), which fits a change in host load during the job rather than random jitter. We therefore read the Qwen savings as upper bounds that a same-node, exclusively allocated, interleaved rerun would need to confirm. Clocks were not logged. AWQ-4 used the GEMM `awq` kernel rather than `awq_marlin`. On A100 the Marlin path is the one vLLM usually selects, and it is typically faster, so every AWQ throughput number is a kernel choice as well as a format property. That choice can contribute to Llama AWQ-4 decoding more slowly than BF16 on Condition A (70.20 vs. 72.34 tok/s). GPTQ-4 timing used `float16` while GPTQ-4 pass@1 used `bfloat16`, so the hybrid ratio joins two configurations. These FP8 numbers are A100 Marlin W8A16, not Hopper W8A8. Peak allocated VRAM is 53.9–56.1 GB, the 0.75 engine pool. Condition A latency is in Appendix Table 19.

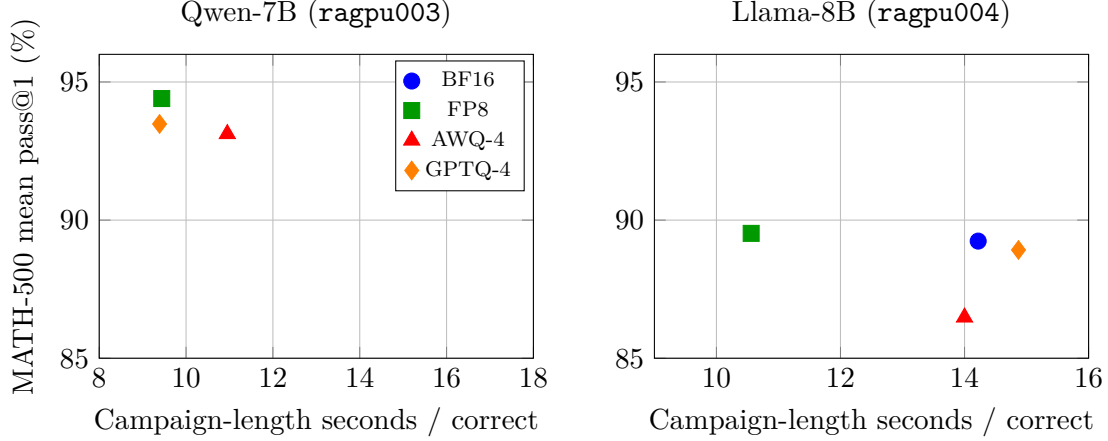


Figure 2: Campaign-length seconds per correct answer versus MATH-500 pass@1. The horizontal axis is full-grid mean tokens divided by the measured Condition B tok/s and by pass@1. There are no timing whiskers: $R = 3$ or 5 does not support a percentile interval, and those repeats do not resample length. Points are not joined. Panels are separate hosts and are not a pooled trade-off plot.

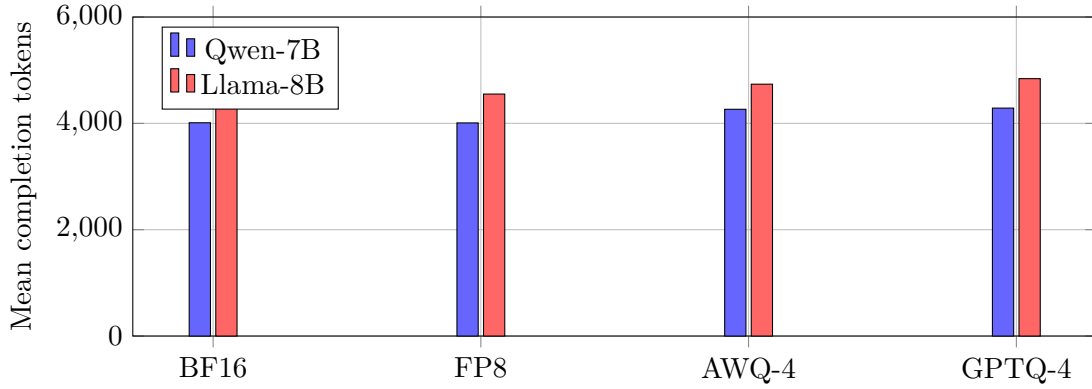


Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand). Bars are point estimates. Item-level 95% intervals on paired token Δ are in Table 8 and are not drawn on this axis.

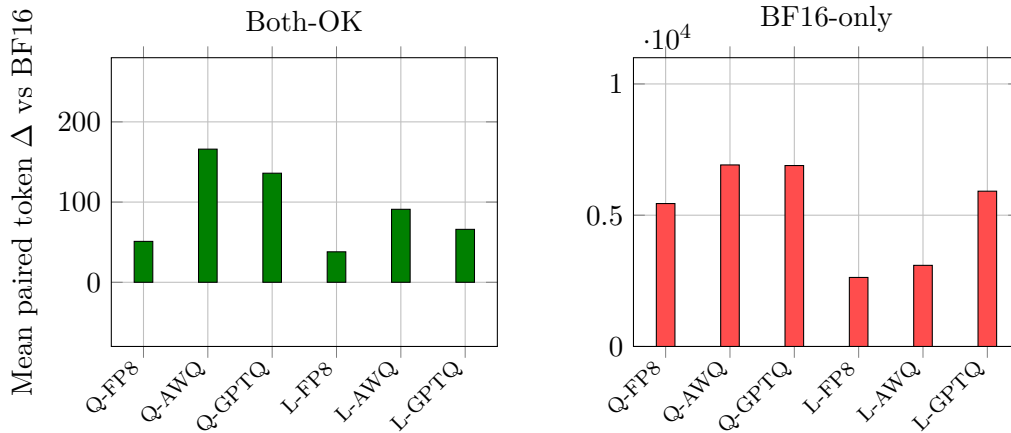


Figure 4: Paired token deltas by correctness, two panels so Both-OK is visible. Left: jointly-correct pairs (Qwen AWQ/GPTQ CIs exclude 0; FP8 and Llama CIs include 0; n in Table 8). Right: BF16-correct / quantized-wrong pairs. Clustered 95% CIs and n are in Table 8.

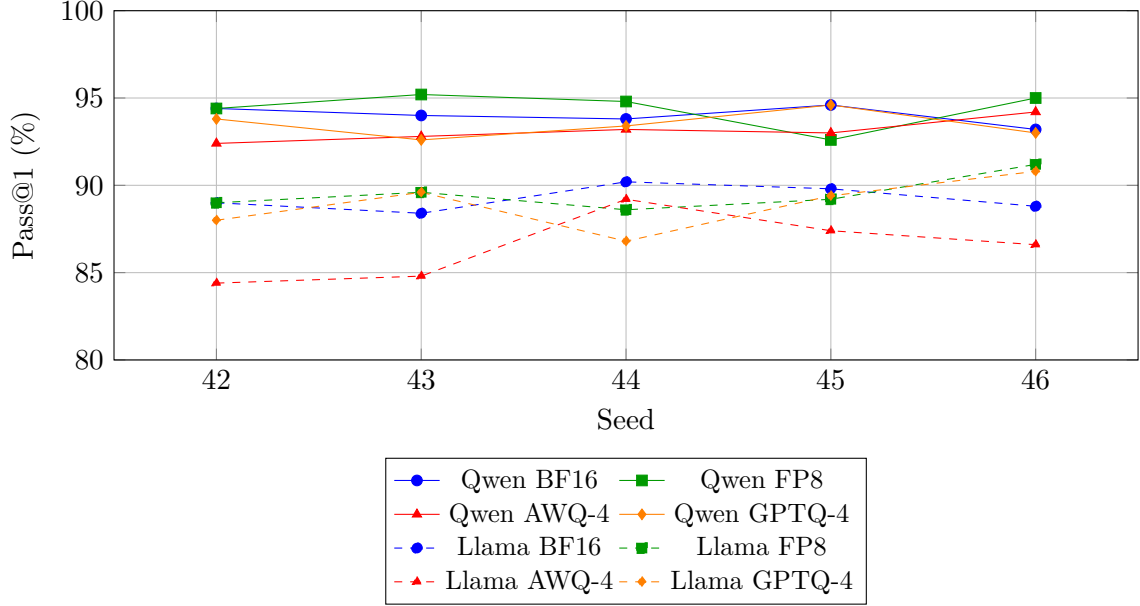


Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells.

Table 15: Campaign-length seconds per correct answer: full-grid mean MATH-500 tokens $\div$ measured tok/s $\div$ pass@1. Tok/s is the confirmation mean and is held fixed. Lower is cheaper. This is a sensitivity to subset length, not a replacement for Table 14.

Model	Fmt	A sec.	A rank	B sec.	B rank
Qwen-7B	BF16	97.16	4	16.89	4
Qwen-7B	FP8	67.93	3	9.44	2
Qwen-7B	AWQ-4	59.57	2	10.95	3
Qwen-7B	GPTQ-4	55.48	1	9.39	1
Llama-8B	BF16	72.13	2	14.22	3
Llama-8B	FP8	64.55	1	10.56	1
Llama-8B	AWQ-4	78.02	3	14.00	2
Llama-8B	GPTQ-4	85.74	4	14.87	4

5 Discussion

Llama AWQ-4 is the accuracy result that survives the item-level intervals on both MATH-500 and GSM8K. maj@5 removes the Qwen AWQ-4 pass@1 gap and about half of the Llama AWQ-4 gap. The BF16-correct length column is positive in every contrast because it includes quantized failures, and a BF16-versus-BF16 placebo reproduces +138 to +235 tokens of it with no quantization (Table 7). The quant-correct column flips that sign in four of six contrasts, and three of those intervals exclude zero (Table 8). A correctness-conditioned length claim should therefore be reported net of its seed placebo.

Five-sample agreement has selective risk at most 0.27%, and a one-sample length rule at matched coverage is higher (Table 12). The gain does not require five samples. Two-sample unanimity already gives 2.2–3.1 times lower risk than the matched length rule at twice the tokens (Table 11).

Subset GPU-seconds and campaign-length seconds do not share a point order. Llama FP8 is cheapest on the campaign-length estimator in every item resample. The subset order is a property of one 20-prompt draw. Timing min–max does not cover that draw. AWQ throughput also depends on the GEMM kernel, and Qwen savings are relative to a BF16 baseline on ragpu003 that looks anomalously slow. A point order among tied cells is not evidence that bit-width has a stable deployment ranking.

6 Limitations

1. **A100 W8A16 fallback.** FP8 checkpoints were executed as Marlin W8A16 on A100, not native W8A8. Numbers do not transfer to Hopper native W8A8.
2. **Measured serving scope.** Throughput and GPU-seconds are measured on balanced MATH-500 subsets, not the full campaign. Repeats share one sampling seed, so cost intervals omit length variance. Condition B elapsed time includes the straggler tail; occupancy time was not stored. $R = 3$ or 5 does not support a stable percentile interval. Eager execution was not compared with CUDA graphs. Qwen and Llama used different hosts. Formats ran sequentially on non-exclusive nodes, and clocks and co-tenant load were not logged. The Qwen BF16 baseline appears slow, so the Qwen savings are upper bounds. The \$1.50/A100-hour rate is a pricing scenario. Peak VRAM is the 0.75 engine pool.
3. **Activation dtype.** AWQ-4 accuracy runs use `float16`. BF16, FP8, and accuracy-campaign GPTQ-4 use `bfloat16`. Serving GPTQ-4 uses `float16`. No BF16-at-`float16` control was run.
4. **Pathology metrology.** The loop detector is identical-word runs ≥ 20 . Phrase-level cycles and `finish_reason` were not saved. Truncation is proxied by near-cap length. A per-item recount needs the campaign prompt text, because compact JSON rows are in pipeline order rather than public release order. GPQA prompts were not reconstructed, because the letter order is a process-RNG draw.
5. **Item order.** Compact JSON rows follow the campaign pipeline order. All paired analyses are within that order and are unaffected: the eight cells agree item by item. Joins against public per-item metadata (difficulty level, prompt length) require the campaign prompt files and are not reported.
6. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be

Table 16: Practitioner summary. $\text{pass}@1$ and $\text{maj}@5$ are MATH-500 percent. Tokens are the campaign mean. A\$ and B\$ are $\hat{C}_{\text{pass}}^{\text{hyb}}$. A sec. and B sec. are the campaign-length sensitivity. * marks the Holm-significant MATH or GSM8K $\text{pass}@1$ contrast.

Model	Fmt	$\text{pass}@1$	$\text{maj}@5$	Tokens	A \$	B \$	A / B sec.
Qwen-7B	BF16	94.00	94.4	4,011	0.0454	0.0060	97.2 / 16.9
Qwen-7B	FP8	94.40	95.0	4,008	0.0373	0.0038	67.9 / 9.4
Qwen-7B	AWQ-4	93.12	94.4	4,265	0.0215	0.0035	59.6 / 11.0
Qwen-7B	GPTQ-4	93.48	94.0	4,287	0.0242	0.0032	55.5 / 9.4
Llama-8B	BF16	89.24	91.2	4,657	0.0310	0.0047	72.1 / 14.2
Llama-8B	FP8	89.52	91.0	4,551	0.0431	0.0043	64.6 / 10.6
Llama-8B	AWQ-4*	86.48	89.8	4,737	0.0431	0.0056	78.0 / 14.0
Llama-8B	GPTQ-4	88.92	91.2	4,841	0.0407	0.0059	85.7 / 14.9

wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property.

- Checkpoint provenance.** Both AWQ-4 checkpoints are `jaki`AJK uploads with unknown calibration. FP8 and GPTQ-4 are RedHatAI artifacts. Format, publisher, and toolchain are collinear. No second AWQ artifact was quantized in this study.
- Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.
- Not evaluated.** This study does not evaluate AIME, LiveCodeBench, 32B models, Hopper native W8A8 FP8, energy measurements, production continuous batching, or probability calibration.
- Reproducibility surface.** Compact JSON, recovered MATH-500 answer strings, confirmation timing, and analysis reports are released. Full chain-of-thought traces are not. Compact JSON omits `finish_reason` and output token IDs. Replay needs an A100-80GB, the listed weights, and `runtime_manifest.json`. The serving records log driver 550.54.14; the accuracy-campaign job logs did not archive the driver. New sensitivities are recomputed by `scripts/analysis/manuscript_sensitivities.py`.
- Contamination.** MATH and GSM8K overlap public pretraining corpora. A paired within-item design reduces but does not eliminate contamination interaction with checkpoint.
- Launcher record.** Effective flags, including `max_model_len=32768`, are reconstructed in `runtime_manifest.json` from the launcher. They are not per-job telemetry. The accuracy-campaign NVIDIA driver was not archived; the serving confirmation logs 550.54.14. Historical harness configs are not the campaign stack.

Threats to validity. Construct: $\text{pass}@1$, $\text{maj}@5$, token length, subset GPU-seconds, and campaign-length seconds are distinct estimands. The hybrid proxy is not Erol et al.’s per-problem Cost-of-Pass. Internal: Conditions A and B differ in subset and regime together; timing repeats do not resample length; Qwen and Llama used different hosts. External: two 7–8B families, one eager A100 stack, and three English benchmarks. Results do not transfer to CUDA-graph serving, Hopper W8A8, or a different publisher’s AWQ artifact.

7 Conclusion

Under this pinned A100 / vLLM 0.7.0 eager stack, the defensible results are one robust accuracy drop, a placebo-sized selection effect in the BF16-correct length estimator, a gold-free agreement signal whose benefit arrives mostly with the second sample, and a subset serving cost that one length draw can reorder. Llama AWQ-4 falls -2.76 pp on MATH-500; maj@5 cuts that gap to about -1.4 pp. A BF16-versus-BF16 placebo reproduces $+138$ to $+235$ tokens of the BF16-correct length increase, and the quant-correct mirror does not keep its sign. Two-sample unanimity has 0.7 – 2.4% selective risk against 2.2 – 5.8% for a matched one-sample length rule; 5/5 agreement reaches at most 0.27% . Llama FP8 is cheapest once campaign length replaces the 20-prompt draw. AWQ-4 results are properties of the `jakiAJK` artifacts, run in `float16` on the GEMM awq kernel. GPQA letter order is not a certified paired contrast.

Reproducibility and Artifact Availability

The released artifact enables verification of reported analyses; reproducing the complete GPU campaign requires equivalent hardware and checkpoint availability.

CPU-reproducible from compact JSON: tables, analysis scripts, and `scripts/analysis/manuscript_sensitivities.py` (see `REPRODUCE.md`). Replay of the 88-run campaign requires an NVIDIA A100-80GB, the listed weights, and `results/reports/runtime_manifest.json`. The accuracy-campaign driver version was not archived (the serving records log 550.54.14). Full chain-of-thought traces are not released.

Code, patches, validation JSON, and analysis reports: [Manish06N/reasoning-compression-lab](#). Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`. Measured serving (confirmation): `results/reports/measured_serving_confirmation/`. First unconstrained timing (provenance only): `results/measured_serving/`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

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CRedit authorship contribution statement

Manish Nandish: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Rajiv Misra:** Conceptualization, Resources, Supervision, Writing – review & editing. **Midhun-chakkaravarthy Janarthanan:** Supervision, Writing – review & editing.

The first author’s dual affiliation (IIT Patna and Lincoln University College) reflects a joint doctoral arrangement. Computational experiments, software, and formal analysis were performed by Manish Nandish. No experimental or software roles are claimed for Midhun-chakkaravarthy Janarthanan.

A Prompt

MATH-500 and GSM8K use the QRM template. Official `inference.py` calls `LightEval` with `use_chat_template=True`:

Table 17: Public checkpoints and revisions used for the 56k campaign.

Cell	Hugging Face ID / revision
Qwen BF16	deepseek-ai/DeepSeek-R1-Distill-Qwen-7B 916b56a44061fd5cd7d6a8fb632557ed4f724f60
Qwen FP8	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic ceb2fcd178d477cd21b92ffb43164c74165a212c
Qwen AWQ-4	jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ 1d082b3ded4e7cefbb6b32cd5e07c7d409832816
Qwen GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16 9559a91ed463eecd64b93a532fe5aa8cc25f2eb
Llama BF16	deepseek-ai/DeepSeek-R1-Distill-Llama-8B 6a6f4aa4197940add57724a7707d069478df56b1
Llama FP8	RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic 5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd
Llama AWQ-4	jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ 57360e84a0e7278c6b3abb453f2dc812b213badb
Llama GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16 faa06ed0a4a5bdab76c4df12376e0c5228c0bdfc

{question}

Please reason step by step, and put your final answer within \boxed{ }.

The prompt-cap recount applies that string through each family’s chat template, including the generation prompt. Saved completions also begin with `<think>`; we did not insert that marker as a separate prefix. GPQA-Diamond uses a different function in QRM’s `lighteval_custom/tasks/reasoning.py`. The prompt asks for a final line `Answer: $LETTER` and inserts the correct choice with `random.randint(0, 3)`. `inference.py` seeds Python’s RNG with the generation seed, then loads the model, then builds prompts. The pipeline’s later `random.seed(1234)` runs only after those prompts exist. Model load can consume RNG draws, and AWQ and BF16 do not take the same init path, so the letter sequence is not certified to match across checkpoints. Stored GPQA rows agree with that template: 4,746 of 4,752 completions contain no `\boxed`, while pass@1 is about 41–53% in every format, including BF16. Extractive match on GPQA is a letter match, not a boxed-answer grade. The QRM comparison row for GPQA in Table 1 is therefore not a paired contrast.

B Checkpoints

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts. This study evaluates those public checkpoints under the dtypes in Table 2 and does not report Red Hat production serving measurements. Dataset revision SHAs are in `results/reports/runtime_manifest.json`.

Serving confirmation protocol and raw JSON live under `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md` and `results/measured_serving_confirmation/raw/`. Aggregates: `measured_serving_confirmation_report.json`. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`, `results/measured_serving/`) is provenance only and is not mixed with confirmation means.

Table 18: Paired coverage Δ versus matched BF16 (percentage points). Intervals that exclude zero are marked *. Selective-accuracy Δ intervals are in the machine-readable modal report.

Contrast	$\geq 3/5$ cov. Δ [95% CI]	$\geq 4/5$ cov. Δ [95% CI]	5/5 cov. Δ [95% CI]
Qwen FP8	+0.6 [−0.8, +2.0]	−0.8 [−2.2, +0.6]	+0.4 [−1.6, +2.4]
Qwen AWQ-4	−0.2 [−1.8, +1.2]	−2.2 [−4.0, −0.6]*	−2.0 [−4.4, +0.2]
Qwen GPTQ-4	−0.6 [−2.0, +0.8]	−0.6 [−2.2, +1.0]	−1.6 [−4.2, +0.8]
Llama FP8	0.0 [−1.8, +1.8]	+0.8 [−1.6, +3.2]	+2.0 [−1.2, +5.2]
Llama AWQ-4	−0.8 [−2.8, +1.2]	−3.2 [−6.0, −0.6]*	−6.0 [−9.4, −2.6]*
Llama GPTQ-4	−0.2 [−2.2, +1.8]	−1.4 [−3.8, +1.0]	−0.8 [−4.2, +2.6]

C Reanalysis

Pathology, the item-level bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10,000$, seed 0.

D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix `23e9ead021111959`; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses $B = 10,000$ problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence = $m_i/5$) are in the JSON report and are not treated as a primary calibration result.

T_5 is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90 T_5 are computed over all 500 items. Tokens per served query and tokens per correct served result use that same 500-item T_5 total in the numerator, so abstained queries are not free.

E Measured serving: latency, microbenchmark, FP8 repeats, and token proxy

Condition A latency detail (Table 19) is sequential decode of a balanced 20-prompt MATH-500 subset. Median latency is the mean of per-run medians. $R = 3$ except Llama AWQ-4 ($R = 5$; CV-triggered). Fixed-token microbenchmark (Table 20): 10 prompts $\times$ 512 tokens, $T = 0.0$, `ignore_eos`. Table 21 lists all five Qwen FP8 Condition B repeats. Table 22 reports mean tokens divided by pass@1. It is not measured wall-clock.

F Neighboring measurement studies

Table 23 is a compact comparison. It is not a claim that this paper replaces those studies.

G Holm-18 sensitivity

Primary multiplicity control is Holm–Bonferroni within each benchmark’s six format contrasts. Table 24 applies Holm jointly to all 18 contrasts as a stricter sensitivity. Llama MATH AWQ-

Table 19: Confirmation single-stream serving (Condition A: 20 sequential prompts). Latency columns are means of per-run median/P90. $\tilde{C}_{\text{pass}}^{\text{hyb}}$ uses campaign MATH-500 pass@1 and a \$1.50/A100-hour scenario. Tok/s is mean $\pm$ sample SD.

Model	Fmt	tok/s	Med. lat (s)	P90 lat (s)	GPU-s/q	$\tilde{C}_{\text{pass}}^{\text{hyb}}$
Qwen-7B	BF16	43.92 $\pm$ 0.04	59.19	244.83	102.51	\$0.0454
Qwen-7B	FP8	62.50 $\pm$ 0.08	43.60	197.01	84.60	\$0.0373
Qwen-7B	AWQ-4	76.89 $\pm$ 0.70	26.47	110.43	48.04	\$0.0215
Qwen-7B	GPTQ-4	82.67 $\pm$ 0.69	34.78	67.51	54.39	\$0.0242
Llama-8B	BF16	72.34 $\pm$ 0.27	32.40	126.10	66.43	\$0.0310
Llama-8B	FP8	78.75 $\pm$ 2.89	38.00	253.55	92.55	\$0.0431
Llama-8B	AWQ-4	70.20 $\pm$ 2.71	40.12	246.14	89.51	\$0.0431
Llama-8B	GPTQ-4	63.49 $\pm$ 0.41	46.31	242.83	86.95	\$0.0407

Table 20: Fixed-token decode microbenchmark (10 prompts $\times$ 512 tokens). Isolates kernel decode speed from trace-length variation. Secondary to task-realistic Conditions A/B.

Model	Fmt	Raw decode tok/s
Qwen-7B	BF16	205.41
Qwen-7B	FP8	437.19
Qwen-7B	AWQ-4	377.73
Qwen-7B	GPTQ-4	401.19
Llama-8B	BF16	362.01
Llama-8B	FP8	392.05
Llama-8B	AWQ-4	344.04
Llama-8B	GPTQ-4	302.54

4 and Llama GSM8K AWQ-4 remain significant. Qwen GPQA AWQ-4 does not (Holm-18 $p = 0.130$, from the $B = 10^6$ bootstrap $p = 0.0081$); it is already borderline under Holm-6.

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Table 21: Qwen-7B FP8 Condition B: five technical repeats with identical aggregate 373,794 output tokens. The tok/s column is a ratio. The mean GPU-s/q of 8.64 implies $3,738/8.64 \approx 432.5$ tok/s, which is not the mean of the five ratios (449.79). Slow: tok/s < 400. Fast: tok/s > 500. The cause of the timing regimes was not identified.

Rep	tok/s	GPU-s/q	$\widehat{C}_{\text{pass}}^{\text{hyb}}$	Regime
1	350.91	10.6522	\$0.0047	slow
2	350.68	10.6591	\$0.0047	slow
3	455.90	8.1990	\$0.0036	mid
4	545.18	6.8563	\$0.0030	fast
5	546.26	6.8427	\$0.0030	fast
Mean of ratios	449.79 $\pm$ 97.53	8.64	\$0.0038	keep all five
Median / [Q1, Q3]	455.90 / [350.91, 545.18]	—	—	—

Table 22: MATH-500 mean tokens divided by pass@1. This is the order a shared throughput would produce. It is not measured wall-clock. Primary cost evidence is Table 14 and Table 15.

Cell	Mean tok.	Tokens / correct
Qwen BF16	4,011.4	4,267
Qwen FP8	4,007.7	4,245
Qwen AWQ-4	4,265.3	4,580
Qwen GPTQ-4	4,287.2	4,586
Llama BF16	4,656.6	5,218
Llama FP8	4,550.8	5,084
Llama AWQ-4	4,736.5	5,477
Llama GPTQ-4	4,840.5	5,443

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Table 23: Neighboring measurement studies. “Engine pinned” means one named serving stack is frozen while checkpoints vary. Length estimands and serving columns are what each paper reports as primary evidence, not an exhaustive inventory.

Study	Models	Bits	Engine pinned	Seeds	Length estimand	Correctness split	Serving / cost	Statistics
This paper	2 R1 distill (7B/8B)	BF16, FPS, AWQ-4, GPTQ-4	vLLM 0.7.0 eager, A100	5 MATH; 3 GSM/GPQA	BoM, paired Δ , both conditionals, seed placebo	2×2 + both mirrors	GPU-c, campaign-length seconds	item-level bootstrap, Holm, TOST
QRM [18]	R1 distill family	W/A/KV incl. W4A16	QRM harness (multi-config)	typically 1	mean tokens	not 2×2	limited	per-cell means
Lian et al. [14]	reasoning PTQ	low-bit PTQ	paper stack	paper-specific	CTIR (ratio of means); BF16-correct split (suppl.)	BF16-correct vs rest (suppl.)	token cost	length ratio
Laffi et al. [19]	1.5B-32B reasoning	aggressive PTQ	paper stack	paper-specific	CuT length; overthinking	abandoned intermediates	logit-penalty Pareto	error-type counts
Kurtic et al. [11]	Llama-3.1 + R1-distill	FPS/INT8/INT4	vLLM, several GPUs	large evals	net CoT-primary	N/A	sync vs continuous batching	accuracy + tok/s
Heigig et al. [8]	Llama/Qwen (SLQ)	3.3-6 bpp	vLLM kernels	3 (GSM)	thinking tokens	TL vs DL fidelity	vLLM throughput	EAR, Wilcoxon
End et al. [9]	general LMs	N/A	N/A	N/A	N/A	N/A	C_{pass} definition	economic
Hochhammer et al. [10]	reasoning LMs	N/A	deliberately unpinned	many	N/A	N/A	N/A	reproducibility

Table 24: Secondary Holm correction over all 18 pass@1 contrasts. Primary inference remains Holm-6 within each benchmark. sig.: Holm-significant at $\alpha = 0.05$; border.: borderline (see Section 3.5); n.s.: not significant.

Bench.	Contrast	p	Holm-6	Holm-18 p	Holm-18
MATH	Llama AWQ-4	< 0.001	sig.	< 0.001	sig.
MATH	Qwen AWQ-4	0.069	n.s.	1.000	n.s.
MATH	Qwen GPTQ-4	0.245	n.s.	1.000	n.s.
MATH	Qwen FP8	0.383	n.s.	1.000	n.s.
MATH	Llama GPTQ-4	0.619	n.s.	1.000	n.s.
MATH	Llama FP8	0.647	n.s.	1.000	n.s.
GSM8K	Llama AWQ-4	0.002	sig.	0.031	sig.
GSM8K	Llama GPTQ-4	0.511	n.s.	1.000	n.s.
GSM8K	Qwen AWQ-4	0.579	n.s.	1.000	n.s.
GSM8K	Qwen GPTQ-4	0.729	n.s.	1.000	n.s.
GSM8K	Llama FP8	0.761	n.s.	1.000	n.s.
GSM8K	Qwen FP8	0.827	n.s.	1.000	n.s.
GPQA	Qwen AWQ-4	0.008	border.	0.130	n.s.
GPQA	Qwen GPTQ-4	0.288	n.s.	1.000	n.s.
GPQA	Llama FP8	0.403	n.s.	1.000	n.s.
GPQA	Llama GPTQ-4	0.573	n.s.	1.000	n.s.
GPQA	Qwen FP8	0.690	n.s.	1.000	n.s.
GPQA	Llama AWQ-4	0.711	n.s.	1.000	n.s.

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